Applied Research Overview

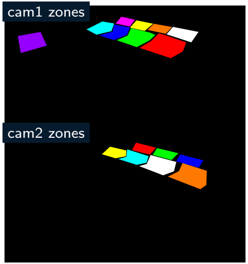
Smart i-LAB AIR1 All-Zones

Room-level occupancy estimation from two cameras and shared environmental sensing

pctiope/smart-i-lab-testbed

May 2026

Problem Motivation



Tracked camera-zone masks define AIR1 coverage.

Research setting

AIR1 asks for room-level occupancy over zones 1–15. The evidence is not one desk, one camera, or one sensor stream: two camera views, per-zone masks, shared AIR-1 readings, SEN55 readings, and readiness gates all shape the result.

- Camera coverage differs by zone and can be missing for a timestamp.
- The dashboard should remain useful before a production model is promoted.
- The model must not borrow Zone 5 power or mmWave assumptions.

Research Gap

What all-zones changes

- A room model must preserve per-zone labels, not collapse the room into one binary stream.
- Long-form `timestamp + zone_id` rows need split, grouping, audit, and display discipline.
- Missing camera coverage should remain unknown, not become vacancy.

Gap addressed here

AIR1 is framed as an all-zones applied-research pipeline: per-zone CV labels, two-camera coverage, shared environmental features, null-preserving labels, and model-readiness gates are evaluated together.

Zones 1–15 | Two cameras | No power/mmWave

Research Questions

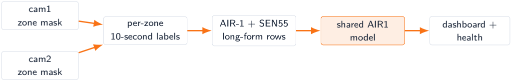
1. Can one shared model represent zones 1–15 from long-form `timestamp + zone_id` rows?
2. Can per-zone CV labels stay null when coverage is missing instead of becoming false negatives?
3. Can the AIR1 dashboard stay useful while inference is gated by production model readiness?

The AIR1 deck keeps Zone 5-specific power, mmWave, and CI/CD scope out of the model narrative except as contrast.

System And Method Overview

Method

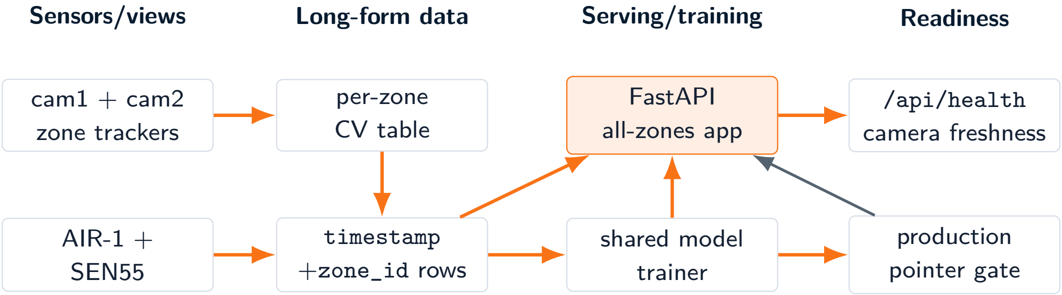
- Use cam1 and cam2 masks to emit per-zone CV counts.
- Aggregate labels into 10-second `timestamp + zone_id` buckets.
- Join AIR-1 and SEN55 features into long-form rows for model zones 1–15.
- Use `occupied` as the target and keep `zone_id` for grouping, audits, splits, and display.
- Gate live inference on a promoted production pointer while keeping camera/dashboard health visible.



Contract boundary

AIR1 model inputs are AIR-1, SEN55, missing indicators, and time features. Power, mmWave, and `zone_id` are excluded from the feature input.

Technical Architecture As Method Evidence



Null-preserving labels | Shared model | Per-zone dashboard | Readiness before inference

Labeling And Feature Contract

Label contract

- Target: `occupied`.
- Key: one row per `timestamp` + `zone_id`.
- Positive rule: per-zone median `occupancy_count >= 1`.
- No matching CV label means `occupied = null`.
- Table 16 is visible camera context but excluded from model zones.

Feature contract

- Valid model zones are 1–15.
- Inputs: AIR-1 environment, SEN55 fields, missing indicators, and time cycles.
- `zone_id` supports grouping, windows, audits, splits, and display only.
- No power or mmWave features are imported from Zone 5.

Evaluation Design And Validation Gates

Offline validation

- Per-zone label coverage and both-class window checks.
- Rolling-calendar splits with hidden blind-test evidence.
- Contract tests for feature columns, masks, labels, and runtime payloads.
- Smoke checks reject wrong target columns and incompatible artifacts.

Runtime validation

- `/api/health` reports readiness, history size, camera statuses, and latest errors.
- Camera freshness remains visible even when inference waits for a production pointer.
- Missing production readiness is a gated state, not a dashboard outage.
- Null label coverage is audited instead of converted to vacancy.

Dashboard Usefulness While Inference Is Gated

Useful before promotion

- Two MJPEG feeds can show whether cam1 and cam2 are fresh.
- Per-zone CV truth can be displayed independently from model probability.
- The health payload explains readiness blockers such as a missing production pointer.

Applied-research implication

AIR1 separates observation quality from model readiness. A dashboard can support data collection and diagnosis before the first promotable all-zones model exists, as long as it does not present missing inference as validated occupancy.

Collect Audit Gate Promote

Limitations And Threats To Validity

Data and label threats

- Camera occlusion and zone-mask coverage can vary by table and view.
- Missing camera coverage creates null labels and reduces usable validation evidence.
- Table 16 exclusion limits model scope to zones 1–15.
- Calendar-window maturity can block strict rolling validation.

Operational threats

- Production pointer absence means inference is not ready, even if collection is alive.
- Sensor/API availability can affect environmental feature freshness.
- Training quality depends on enough occupied and unoccupied windows per split.
- This deck intentionally excludes Zone 5 deployment claims from AIR1 scope.

Research Roadmap And Source Anchors

Roadmap

- Mature per-zone label coverage before treating all-zones inference as production-ready.
- Promote only after strict-window evidence, smoke checks, and contract tests pass.
- Keep camera freshness, null label rates, and production pointer state visible in operations.

Topic	Primary source paths
Pipeline contract	`air1/.../PIPELINE.md`
Feature columns	`air1/.../feature_contract.py`
CV labels and masks	`air1/.../rtsp_zone_tracker.py`, `air1/.../masks/`
Training/promotion	`air1/.../retrain_once.py`, `air1/.../promote_model.py`
Runtime readiness	`air1/.../web_app/`, `air1/.../tests/`

Closing point: AIR1 is an all-zones, two-camera occupancy study whose null labels and readiness gates are part of the method.